MicroROM-NPU

Microinstruction Format & Field Definitions

Design reference • 48-bit vertical microinstruction • Draft specification v0.1

Purpose. This document defines the proposed 48-bit internal control word for the MicroROM-NPU project. It is an internal microarchitecture format, not a RISC-V machine-instruction encoding. Architectural instructions are decoded and dispatched to micro-routines stored in MicroROM.

Status note. This is a draft design specification. Field positions and candidate encodings are documented here, but the full control logic and all encoded ROM words must be validated in SystemVerilog simulation before use. Reserved codes must not be treated as implemented behavior.

1. Microinstruction format

Every microinstruction is exactly 48 bits wide. The bit positions below are inclusive, with bit 47 as the most significant bit.

Field	Width	Bit positions	Primary role
UOP	5	[47:43]	Selects the micro-operation performed by the datapath.
SRC1_SEL	6	[42:37]	Selects the first source operand.
SRC2_SEL	6	[36:31]	Selects the second source operand.
DST_SEL	6	[30:25]	Selects the destination register/bank entry.
NEGATE	1	[24]	Optionally negates SRC2 for supported operations; distinct from UOP=NEG.
IMM_SEL	3	[23:21]	Selects an immediate or constant used by an immediate operation.
MEM_CTRL	3	[20:18]	Selects the memory access type.
COND	3	[17:15]	Selects a condition evaluated by conditional microsequencing.
NEXT	3	[14:12]	Selects the microsequencer's next-address mode.
NEXT_ADDR	6	[11:6]	Absolute target microaddress for jump/conditional modes.
PC_CTRL	2	[5:4]	Controls the architectural program counter (PC).
FLAGS_CTRL	2	[3:2]	Controls updates to comparison/condition state.
RESERVED	2	[1:0]	Reserved for future use; set to zero in this draft.

Width check: 5 + 6 + 6 + 6 + 1 + 3 + 3 + 3 + 3 + 6 + 2 + 2 + 2 = 48 bits.

Bit-field layout (MSB → LSB)

UOP	SRC1	SRC2	DST	NEG	IMM	MEM	COND	NEXT	NEXT_ADDR	PC	FLAGS	RSVD
5	6	6	6	1	3	3	3	3	6	2	2	2

2. Meaning of each field

2.1 UOP — Micro-operation (5 bits)

Code	Name	Defined behavior
00000	NOP	No datapath write.
00001	MOV	DEST ← SRC1.
00010	ADD	DEST ← SRC1 + SRC2.
00011	ADDI	DEST ← SRC1 + selected immediate.
00100	NEG	DEST ← −SRC1 using two's-complement negation.
00101	CMP	Capture comparison information for SRC1 versus SRC2; do not write a general-purpose destination.
00110	XOR_SIGN	DEST ← sign(SRC1) XOR sign(SRC2), normalized to 0 or 1.
00111	CLR	DEST ← 0.
01000	LOAD	Load selected memory data into DEST; sign extension applies for signed-byte reads.
01001	STORE	Store SRC1 to the selected memory address.
01010	DECODE	Decode the fetched architectural instruction.
01011	PC_UPDATE	Apply the architectural PC action selected by PC_CTRL.
01100	WRITE_RD	Write a result to the architectural destination register selected by instruction decode.
01101	FETCH	Fetch the next architectural instruction.
01110	RETURN	No datapath write; use NEXT=RETURN for micro-return sequencing.
01111–11111	Reserved	Not implemented in this draft.

2.2 SRC1_SEL, SRC2_SEL, DST_SEL — Register selectors (6 bits each)

The selector encoding allows the control word to choose an architectural register or an internal register. This is a shared selector namespace only: the architectural register file and the internal register bank remain separate physical/logical state.

Selector	Selected register
000000–011111	Architectural x0–x31 (selector value equals register number).
100000 (32)	Internal A — first multiplication operand / activation magnitude during multiply routine.
100001 (33)	Internal B — second multiplication operand / weight magnitude during multiply routine.
100010 (34)	Internal TMP — temporary scratch register.
100011 (35)	Internal ACC — signed 32-bit accumulated sum, including bias.
100100 (36)	Internal PROD — signed product/result temporary.
100101 (37)	Internal CTR — loop counter for repeated-add multiplication.
100110 (38)	Internal MDR — memory data register.
100111 (39)	Internal MAR — memory address register.
101000 (40)	Internal SIGN — 0/1 product-sign indicator.
101001–111111	Reserved selectors.

Architectural x0: reads as zero and ignores writes. Internal selectors must not alias architectural register indices.

2.3 NEGATE — Optional second-operand negation (1 bit)

Value	Meaning
0	Use SRC2 unchanged.
1	For operations explicitly supporting it (initially ADD), use −SRC2. Example: DEST ← SRC1 + (−SRC2).

UOP=NEG is a separate operation that computes DEST ← −SRC1. Do not apply NEGATE to UOP=NEG; this avoids ambiguous or double-negation behavior.

2.4 IMM_SEL — Immediate/constant selector (3 bits)

Code	Meaning
000	Constant 0.
001	Sign-extended constant −1 (used to decrement CTR).
010	Decoded architectural I-type immediate.
011	Decoded architectural S-type immediate.
100	Decoded architectural B-type branch immediate.
101	Decoded architectural J-type jump immediate.
110	Constant +1.
111	Reserved.

The architectural decoder must form and sign-extend immediates according to the corresponding instruction format. IMM_SEL chooses among these already-defined sources; it does not itself perform instruction decoding.

2.5 MEM_CTRL — Memory access control (3 bits)

Code	Meaning
000	No memory access.
001	Read signed byte; sign-extend to 32 bits.
010	Read 32-bit word.
011	Write 32-bit word.
100–111	Reserved.

Address generation, access alignment, and exception behavior must be defined by the implemented architectural subset. These details must be tested before claiming RISC-V compliance.

2. Meaning of each field (continued)

2.6 COND — Conditional test (3 bits)

Code	Meaning
000	EQ — comparison operands are equal.
001	NE — comparison operands are not equal.
010	GE — signed SRC1 ≥ signed SRC2.
011	LT — signed SRC1 < signed SRC2.
100	ZERO — selected/latched test result is zero; exact source must be fixed in implementation.
101	NONZERO — selected/latched test result is nonzero; exact source must be fixed in implementation.
110–111	Reserved.

Condition-state rule: CMP captures the comparison information; NEXT=COND evaluates COND using that captured state. The implementation must explicitly define how ZERO/NONZERO obtain their tested value, or reserve these two codes until that source is specified.

2.7 NEXT — Microsequencer mode (3 bits)

Code	Mode	Meaning
000	SEQ	μPC ← μPC + 1.
001	JUMP	μPC ← NEXT_ADDR.
010	COND	If the selected condition is true, μPC ← NEXT_ADDR; otherwise μPC ← μPC + 1.
011	DISPATCH	Select a micro-routine entry address from the decoded architectural instruction.
100	RETURN	Restore μPC from the saved micro-return register μRET.
101–111	Reserved	Not implemented in this draft.

NEXT_ADDR is a 6-bit absolute microaddress in this draft. A single μRET register supports only one outstanding micro-routine call; nested calls require a stack or another explicit mechanism.

2.8 NEXT_ADDR — Microaddress target (6 bits)

Provides the target address for NEXT=JUMP and the taken path of NEXT=COND. For NEXT=SEQ, NEXT=DISPATCH, and NEXT=RETURN, the field is ignored. A 6-bit field can address 64 microinstructions (0–63).

2.9 PC_CTRL — Architectural program counter control (2 bits)

Code	Meaning
00	Hold architectural PC.
01	Advance PC by the current architectural instruction length (2 or 4 bytes for the planned RV32C-capable subset).
10	Load decoded branch/jump target into PC.
11	Reserved.

PC and μPC are distinct. PC tracks architectural program execution; μPC tracks the internal microprogram. For a taken branch or jump, the decoder/control logic must supply the correctly formed target.

2.10 FLAGS_CTRL — Comparison-state update (2 bits)

Code	Meaning
00	Preserve comparison state.
01	Update equality/zero comparison information.
10	Update signed comparison information.
11	Reserved.

The implementation may latch comparison operands and evaluate EQ/NE/GE/LT from them, or latch explicit comparison outcomes. Whichever method is selected must be used consistently by CMP and the conditional microsequencer.

2.11 RESERVED — Reserved bits (2 bits)

Bits [1:0] are set to 00 in this draft. Future versions may assign them new behavior, but a decoder should not infer any behavior from nonzero reserved values until the format is deliberately revised.

3. Design conventions and validation checklist

• Internal vs architectural state: keep the architectural x0–x31 register file separate from A, B, TMP, ACC, PROD, CTR, MDR, MAR, and SIGN.

• Signedness: A and B are assumed to contain sign-extended INT8 values on entry to the multiply routine; ACC is signed 32-bit.

• Negation: NEG performs two's-complement negation (~value + 1); NEGATE is a separate optional SRC2 transformation for supported ALU operations.

• Comparison timing: comparison state must be captured before a later microinstruction consumes it for branching.

• PC timing: architectural PC updates must use the instruction's actual length (2 or 4 bytes) where sequential advancement is selected.

• Microsequencing: define μRET save/restore behavior and dispatch-address mapping before running the full instruction set.

• ROM words: generate words using a single canonical packer or SystemVerilog struct/function; verify all fields in simulation rather than hand-trusting hexadecimal constants.

• Scope: this is a proposed purpose-built implementation of a selected RISC-V instruction subset, not a claim of full RV32I/RV32C compliance.

Suggested next implementation artifact: a SystemVerilog package containing field constants, selector constants, a packed microinstruction type or pack function, and assertions that reserved bits are zero and branch targets are in range.